

remain available to the user until the session is terminated.

Tables created in temp space will survive a restart. A query example is given below:

```
CREATE DATABASE new_db FROM existing_db
AS PERMANENT = 2000000, SPOOL =
5000000, TEMP = 2000000;
```

A new database must be created from an existing database. Permanent space can be used for tables in the database. Spool space is allocated for the maximum amount of workspace available for requests. Temp space is allocated for temp tables in the database.

## Types of tables in Teradata

**Permanent tables:** These remain in the system until they are dropped. The table definition is stored in the data dictionary. Data and structure can be shared across multiple sessions and users. Data is stored in the permanent space, and 'collect statistics' is supported. Indexes can be created. COMPRESS columns, as well as DEFAULT and TITLE clauses are supported. Partition primary index (PPI) is also supported. If the primary index clause is not defined in the 'Create table' statement, then Teradata will create the first column as primary by default.

**Global temporary tables (GTT):** This is a kind of temporary table. The table definition is stored in the data dictionary. It requires a minimum of 512 bytes from the permanent space to store table definitions in the data dictionary. The structure can be shared across multiple users but data will remain private to the session (and its user). Data is stored in the temporary space. Data will be purged for that session once the session ends. 'Collect statistics' is supported. Indexes can be created. COMPRESS columns, as well as DEFAULT and TITLE clauses are supported. PPI is also supported.

This table can be identified from the data dictionary table (*dbc.tables*) using the 'CommitOpt' column. If its value is 'D' (on *commit Delete* rows) or 'P' (on *commit Preserve* rows), then it's a GTT. If the primary index clause is not specified while creating the table, then Teradata will create the first column as primary by default. One session can generate up to 2000 global temporary tables at a time.

**Volatile tables:** These are created in the spool space for temporary use, and their life span extends for the duration of the session only. The table definition is not stored in the data dictionary. Structure and data is private to the session and its user. Data is stored in the spool space. The table gets dropped once the session ends, i.e., when 'login again...no volatile tables' shows up. 'Collect statistics' is supported. Indexes cannot be created. The COMPRESS column is supported, but the DEFAULT and TITLE clauses are not supported. PPI is supported. If the primary index clause is not specified while creating the table, then Teradata will create the first column as primary by default.

```
ON COMMIT DELETE ROWS (Default)
ON COMMIT PRESERVE ROWS (If we want to
retain rows)
```

**Derived tables:** This temporary table is derived from one or more other tables as the result of a sub-query. Derived tables are local to the query and exist only for the duration of the query. The table is

automatically deleted once the query is done. Data is stored in the spool space. Table definition is not stored in the data dictionary. The first column in a derived table acts like a PI column for it.

## Properties of Teradata tables

- **Fallback:** This protects data by storing a second copy of each row of a table on a different AMP in the same cluster. If an AMP fails, the system accesses the fallback rows to meet requests.
- **Journals:** BEFORE JOURNAL holds the image of impacted rows before any changes are made. AFTER JOURNAL holds the image of affected rows after changes are done. In DUAL BEFORE/AFTER JOURNAL, two images are taken and are stored in two separate AMPs.
- **Checksum:** This detects some forms of lost writes. A lost write is a write of a file system block that received successful status on completion, but the write either never actually occurred or was written to an incorrect location. As a result, subsequent reads of the same block return the old data. NONE, LOW, MEDIUM, and HIGH levels are available in the checksum.

Teradata is a powerful RDBMS and can help in transforming how businesses work. We have just discussed its basics in this article and a lot still needs to be unearthed. We hope you find this basic information useful to explore the features and capabilities of Teradata further. **END** 

### By: Saurabh Kumar and Surabhi Kumari

**Saurabh Kumar** is a senior data architect in the data science division of American Express. He has worked in several domains like MIS, data analytics, business intelligence, fraud detection, and data anonymization.

**Surabhi Kumari** is well experienced in the field of IT. She is a quality analyst at a leading IT company. Her areas of interest are software testing with automation, APIs, and SQL.